CLAIP: Dual Corroboration and Segment-Aligned Contrastive Learning for Domain Ontology Construction from Meeting Video, Toward Meeting Intelligence

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Abstract

Specialized engineering organizations accumulate much of their domain expertise in recorded design-review meeting video, yet this content rarely feeds back into a reusable, independently verified domain ontology. We pursue this capability – deriving and corroborating structured domain knowledge directly from a meeting's own audio, screen-share, and speech content, toward what we call *Meeting Intelligence* – using a bridge-repair design-review meeting as a concrete case study. Prior work on domain-dictionary correction for Japanese construction-meeting ASR took a first step in this direction, corroborating candidate terms against a single, independently verified source – a static damage-cause ontology – mined either in bulk or from the meeting's own corrected audio transcript *after* speech recognition had already run. This paper extends that pipeline with a second, independent corroboration source available *before* speech recognition runs: optical character recognition (OCR) and image captioning applied to video frames sampled from the meeting's own screen-share content (slides, sticky-note review comments), merged into the domain dictionary so that the automatic speech recognizer's own vocabulary bias already reflects what was visibly written on screen. We further build a shared three-modality (audio, image, text) embedding space via contrastive (CLAIP-style) projection-head training over the meeting's own utterance-aligned data, as a first step toward cross-modal meeting search. We compare two frame-sampling strategies for this image branch on a real 44.9-minute Google Meet recording (914 transcribed utterances, 6 speakers): fixed 10-second-interval sampling and diarization-utterance-segment-driven dense sampling. Under a controlled, fixed-reference-vocabulary evaluation, segment-driven sampling nearly triples kept, de-duplicated frames (92 to 264) and improves candidate-term coverage by 69% (55 to 93 terms, 42 entirely new), while guaranteeing at least one contrastive training triple per transcribed utterance and increasing total triples more than tenfold (90 to 961; all 914 utterances covered, 43 contributing more than one triple because the shared screen changed mid-utterance). This denser, utterance-aligned training data also nearly doubles a qualitative cross-modal retrieval sanity check (2/7 to 4/7 known domain terms correctly retrieved). We additionally report two previously unreported OCR-specific candidate-extraction failure modes – reviewer-name/content concatenation and adjacent sticky-note text duplication – with lightweight, non-LLM mitigations, and an empirical training-stability finding: increasing contrastive-triple count by an order of magnitude induces late-training loss instability under fixed-learning-rate, full-batch Adam, mitigated here by best-checkpoint retention. Motivated by a specific failure case – a Google Meet standby tile (black background, centered name/photo) entering a triple in place of genuine shared-screen content, which should weaken the image-paired losses while leaving the audio-text pair unaffected – we introduce a per-modality-pair loss-weighting hyperparameter and test one hand-chosen setting (down-weighting the two image-paired terms, up-weighting audio-text) in a seed-matched controlled comparison: it sharpens the audio-text pair by 19.8% (rescored on a common, unweighted basis) without measurably degrading either image-paired term, reduces the number of late-training loss spikes from two to one, and leaves the 7-query retrieval hit rate unchanged (4/7), suggesting the weighting is a safe, if modest, refinement

rather than a generalizable fix. We report all results candidly as a single-video case study: the retrieval numbers are a qualitative proof of concept, not a statistically validated benchmark. Taken together, these results are a concrete, if preliminary, step toward Meeting Intelligence: showing that a single bridge-repair design-review recording's own video and audio can already corroborate and expand a verified domain ontology, a mechanism we intend to generalize across construction's other specialized engineering departments in future work.

Keywords: Automatic Speech Recognition, Multimodal Corroboration, Contrastive Representation Learning, Optical Character Recognition, Domain Dictionary Correction, Speaker Diarization, Ontology Construction, Meeting Intelligence.

1 Introduction

1.1 Background and Motivation

Infrastructure-engineering organizations hold much of their domain expertise not in written documents but in the spoken and visually shared content of recorded design-review meetings – videos that are rarely mined back into a reusable, verified domain ontology once the meeting ends. This paper's overarching goal is *Meeting Intelligence*: constructing and continuously corroborating a specialized-domain ontology directly from an organization's own meeting video, using a real bridge-repair external design-review recording as our concrete case study throughout.

A companion paper [1] (hereafter, *the base paper*) presented an on-premise pipeline for Japanese construction-meeting speech recognition that corrects domain-specific vocabulary errors via a morphology-aware, two-pass dictionary-correction algorithm, and expands that dictionary's coverage by corroborating candidate terms against an independently verified damage-cause ontology [2], either in bulk (meeting-independent) or per-meeting, by mining the pipeline's own corrected transcript *after* automatic speech recognition (ASR) has already run. That paper's own concluding future work identified audio-visual capture (Microsoft Teams, Google Meet, Zoom) as an open engineering direction, and noted that the raw ASR transcript alone under-exploits the lived engineering experience voiced in review meetings.

Construction-design review meetings, however, are rarely audio-only in substance: participants routinely share their screen to walk through design drawings, inspection photographs, and document-viewer sessions annotated with stacked sticky-note review comments, each naming a specific structural member, damage mode, or repair method in written form. This visual channel is a second, independent source of exactly the domain vocabulary that the base paper's dictionary-correction algorithm is trying to recover from audio alone – and, critically, this visual content is available to the pipeline *before* ASR is run on a given recording, unlike the base paper's per-meeting audio-side extraction, which is necessarily post-hoc. This paper designs and evaluates a mechanism that corroborates candidate terms using this screen-share content, merges it into the dictionary before ASR runs, and, following the project's own Concept Note's CLAIP (Contrastive Language-Audio-Image Pre-training) vision, projects audio, image, and text representations of the meeting into one shared embedding space via contrastive learning, as a foundation for future cross-modal meeting search.

1.2 Problem Statement and Challenges

Extending dictionary corroboration and contrastive representation learning to a meeting's own screen-share frames raises three challenges addressed in this paper: (a) how densely frames should be sampled from the source video – a fixed wall-clock interval is simple but samples independently of the speech it will be aligned with, and may both miss slide transitions that happen between sample points and over-sample long stretches of an unchanging slide; (b) OCR text extracted from a screenshot of stacked review comments has none of natural language's word-boundary and particle/honorific cues that the base paper's audio-side candidate-extraction mitigations rely on, so OCR-specific candidate-extraction failure modes must be identified and mitigated on their own terms; and (c) aligning each transcribed utterance with the frame(s) shown while it was spoken, and training a shared embedding space over the resulting (image, audio, text) triples, must both guarantee adequate training-data density and remain numerically stable as that density grows by an order of magnitude.

1.3 Contributions

This paper makes the following contributions:

1. **Dual corroboration.** We extend the base paper's ontology-corroboration dictionary-expansion mechanism with a second, independent in-meeting evidence source – screen-share frame OCR – merged into the dictionary *before* ASR executes, so the recognizer's own vocabulary bias (its `initial_prompt`) already reflects what was visibly

written on screen, unlike the base paper's audio-side per-meeting mechanism, which is necessarily post-hoc.

2. **A controlled comparison of two frame-sampling strategies.** We compare fixed 10-second-interval sampling against diarization-utterance-segment-driven dense sampling under a fixed-reference-vocabulary evaluation designed specifically to isolate the sampling method's effect from confounding dictionary-state differences, finding the latter improves candidate-term coverage by 69% (55 to 93 of 183 reference terms, 42 entirely new) and nearly triples kept, de-duplicated frames (92 to 264).
3. **A segment-centric contrastive-triple construction algorithm** that guarantees at least one (image, audio, text) training triple per transcribed utterance (914/914 covered) and one additional triple for each distinct image shown if the shared screen changed mid-utterance (43 utterances), replacing a looser frame-centric, time-window heuristic and increasing total training triples more than tenfold (90 to 961).
4. **Two previously unreported OCR-specific candidate-extraction failure modes** – reviewer-name/content concatenation and adjacent sticky-note text duplication, both arising from OCR text's lack of natural-language word-boundary cues – together with lightweight, part-of-speech-based and string-level mitigations validated on real extracted candidates.
5. **An empirical contrastive-training-stability finding**: scaling triple count by an order of magnitude (90 to 961) induces late-training loss instability under fixed-learning-rate, full-batch Adam optimization of InfoNCE projection heads, which we mitigate with best-checkpoint retention rather than learning-rate scheduling, reported as an engineering finding for small-scale, single-recording contrastive learning.
6. **A qualitative cross-modal retrieval case study**, reported candidly as a single-video proof of concept rather than a statistically validated benchmark, showing the segment-centric design nearly doubles a 7-query sanity-check hit rate (2/7 to 4/7) against known high-confidence domain terms.
7. **A per-modality-pair loss-weighting ablation** motivated by the standby-frame case (§4.7): in a seed-matched controlled comparison, down-weighting the image-paired InfoNCE terms and up-weighting the audio-text term sharpens the audio-text pair by 19.8% without degrading either image-paired term, halves the observed late-training loss-spike count, and leaves the retrieval hit rate unchanged – evidence, on this single recording and seed, that the weighting is a safe refinement rather than a generalizable fix.

1.4 Paper Organization

§2 surveys related work in cross-modal contrastive pretraining, document/slide OCR, multilingual text embeddings, and ontology construction and Meeting Intelligence, and positions this paper relative to the base paper. §3 extends the base paper's pipeline formalization with the image branch and the shared embedding space. §4 details the methodology. §5 and §6 report the experimental setup and results. §7 discusses implications and limitations, and §8 concludes.

2 Related Work

2.1 Cross-Modal Contrastive Pretraining

CLIP [3] established that a shared embedding space between image and text encoders, trained with a symmetric InfoNCE contrastive objective [4] over large-scale noisy image-text pairs, enables strong zero-shot transfer; CLAP [5] applies the same recipe to audio-text pairs. The CLAIP design explored in this paper – jointly projecting audio, image, and text representations of the *same* meeting into one space – combines both lines of work into three bilinear objectives (image-text, audio-text, image-audio) over a single recording's own utterance-aligned data, rather than a large external paired corpus; we discuss the resulting small-sample regime's distinct optimization behavior in §7.3. BLIP [6] generates natural-language image captions via a vision-language encoder-decoder, which we use (§4) for frames whose dominant content is a photograph or diagram rather than a document screenshot; rinna's Japanese CLIP release, based on a ViT-B/16 image encoder and a 12-layer Japanese text encoder [7], provides the domain-appropriate (Japanese-capable) image-text branch of our shared space.

2.2 Document and Slide OCR

EasyOCR [8] combines a CRAFT-style text-detection network with a CRNN-style recognizer, supporting Japanese and over 80 other languages without requiring a separately installed native binary (unlike Tesseract) or a heavier deep-learning framework dependency

beyond PyTorch (unlike PaddleOCR, which we considered and rejected, §4). Unlike the general-purpose image captioning of §2's preceding subsection, OCR targets the *exact* written characters on screen, which matters directly for our setting: a design-review sticky note reading a specific structural-member name is a much stronger domain-vocabulary signal than a natural-language scene description of the same screenshot (§4).

2.3 Multilingual Text Embeddings

For the text branch of the shared embedding space, we use multilingual-E5 [9], a Sentence-BERT-family [10] embedding model trained with weakly-supervised contrastive pretraining across 100+ languages, via the `sentence-transformers` library, chosen for its balance of Japanese-language coverage and parameter count (base, not large) relative to the 16 GB consumer-GPU budget shared with the other encoders in our stack (§5).

2.4 Ontology Construction and Meeting Intelligence

Classical ontology learning from text progressed from shallow, pattern- and statistics-based term/relation extraction to deep-learning sequence models [11], and more recently to large-language-model-driven extraction, schema induction, and knowledge fusion [12, 13]; multimodal knowledge-graph construction further fuses image, text, and other modalities into a single graph [14]. Enterprise-focused work applies this machinery directly to organizational communication: an LLM-driven, activity-centric knowledge graph unifying emails, calendars, chats, and meeting-preparation artifacts for expertise discovery and analytics [15], and a real-world multimodal meeting dataset and benchmark for long-form transcription and summarization that fuses speech, vision, and text cues [16], both of which we take as evidence that *Meeting Intelligence* – deriving reusable, structured domain knowledge directly from an organization's own recorded meetings – is an active and recognized goal distinct from meeting summarization alone. Unlike these text- or synthetic-corpus-centric knowledge-graph-construction pipelines, this paper corroborates a narrow, hand-verified domain dictionary directly from a single meeting's own screen-share and audio content, trading construction generality for verifiability on exactly the vocabulary a design-review team actually used.

2.5 Positioning

This paper is a direct methodological extension of the base paper [1], which remains the source of the underlying speaker-diarization, ASR, and morphology-aware dictionary-correction mechanisms (§4 reuses its Algorithm 1 and dictionary-expansion equation unchanged). We do not re-evaluate those base mechanisms here; this paper's contribution is the new image branch, dual corroboration, and the CLAIP shared embedding space layered on top, evaluated on the same case-study recording the base paper designates M9.

3 Problem Formulation

We adopt the base paper's notation unchanged (preprocessing p, diarization D, segment-wise ASR T, domain-dictionary correction C_φ, and the dictionary-expansion operators E_{ont}, E_{meet} defined there), and extend its dictionary-expansion equation ($\varphi \leftarrow \varphi_0 \cup \varphi_{\text{ont}} \cup \varphi_{\text{meet}}(g_{\varphi_0}(a))$) with one additional expansion operator and a parallel shared-embedding-space formalization.

Frame extraction. Let $F(a) \to \{f_1, \ldots, f_P\}$ denote a sampling function that extracts a set of video frames from the raw recording a, each f_p carrying a timestamp $\text{ts}(f_p)$. We study two instantiations of F in this paper: F_{int}, which samples at a fixed wall-clock interval independent of $D(w)$, and F_{seg}, which samples one or more frames per diarization segment in $D_{\text{excl}}(w)$ (§4). Critically, both instantiations depend only on a (and, for F_{seg}, on the audio-only diarization $D_{\text{excl}}(w)$) – neither depends on the ASR output $T(\cdot)$ – so $F(a)$ is always computable *before* segment-wise ASR runs.

De-duplication and resolution. A perceptual-hash de-duplication pass $\text{dedup}(\{f_p\}) \to \{(f_p, \text{kept}_p, \rho_p)\}$ marks each frame $\text{kept}_p \in \{0, 1\}$ and assigns a *represents* pointer ρ_p: $\rho_p = p$ if $\text{kept}_p = 1$, otherwise ρ_p is the index of the nearest preceding kept frame with the same (unchanged) visible content. Only kept frames are passed to OCR and captioning (§4); ρ_p lets every candidate frame, kept or not, be resolved back to the one frame whose text was actually extracted.

OCR and captioning. $O(f_p) \to x_p^{\text{ocr}}$ and $\text{Cap}(f_p) \to x_p^{\text{cap}}$ extract on-screen text and a natural-language caption, respectively, for each kept frame; their concatenation $x_p \triangleq x_p^{\text{ocr}} \,\|\, x_p^{\text{cap}}$ is this frame's *unified semantic text*.

Frame-derived vocabulary expansion. Analogously to E_{ont}, we define $E_{\text{frame}}\big(\{x_p^{\text{ocr}}\}\big) \to \varphi_{\text{frame}}$, which mines the kept frames' OCR text for can-

didate terms via the same SudachiPy segmentation as $E_{\text{ont}}/E_{\text{meet}}$ (§4), classifies their relevance with the local LLM exactly as E_{meet} does (unlike E_{ont}'s candidates, a term appearing on one meeting's own screen is not already confirmed domain vocabulary by construction), and corroborates each candidate's resolved canonical $\hat{m}$ against *both* independently verified sources:

$$\text{corrob}(m) \triangleq \mathbb{1}\big[\hat{m} \in \text{dom}(\varphi_{\text{ont}}) \cup \text{dom}(\varphi_{\text{frame}})\big], \quad (1)$$

recording which source(s) fired, $\text{src}(m) \in \{\text{ont}, \text{frame}, \text{both}, \emptyset\}$, generalizing the base paper's single-source Equation (corrob) to this paper's *dual* corroboration. The dictionary used by a given run becomes

$$\varphi \leftarrow \varphi_0 \cup \varphi_{\text{ont}} \cup \varphi_{\text{frame}}\big(F(a)\big) \cup \varphi_{\text{meet}}\big(g_{\varphi'}(a)\big), \quad (2)$$

where $\varphi' \triangleq \varphi_0 \cup \varphi_{\text{ont}} \cup \varphi_{\text{frame}}(F(a))$. Because φ_{frame} depends only on $F(a)$, it is available before $g_{\varphi'}(a)$ (hence before $T(\cdot)$) is computed, letting the newly discovered screen-share vocabulary bias Whisper's own `initial_prompt` *within the same pipeline run* – the central distinction from φ_{meet}, which can only ever expand the dictionary for a *future* run on the same recording.

Shared embedding space. Let $f_{\text{img}}, f_{\text{aud}}, f_{\text{txt}}$ be frozen, pretrained encoders (§4) mapping a kept frame, an audio segment, and a text string to fixed-size embeddings, respectively. For a set of N aligned triples $\{(f_{\text{img}}(\cdot), f_{\text{aud}}(\cdot), f_{\text{txt}}(\cdot))_i\}_{i=1}^{N}$ (§4 defines two alignment strategies, frame-centric and segment-centric), three learned linear projection heads $\pi_{\text{img}}, \pi_{\text{aud}}, \pi_{\text{txt}} : \mathbb{R}^{d_k} \to \mathbb{R}^d$ are trained to maximize agreement between L2-normalized projections of matched pairs via a symmetric InfoNCE objective (Equation 4, §4), summed over all three modality pairs. We study how the construction of the triple set N – in particular, whether it is built per-frame or per-utterance-segment – affects both N's size and the trained space's cross-modal retrieval quality (§6).

4 Methodology

4.1 System Architecture

Figure 1 extends the base paper's pipeline (its Figure 1) with the image branch and shared-embedding stages introduced in this paper. The image branch runs in a separate Python virtual environment from the audio/dictionary branch (§5), orchestrated via subprocess calls so GPU memory for each stage's models can be released before the next stage loads (the same resource-management discipline the base paper applies across diarization and ASR).

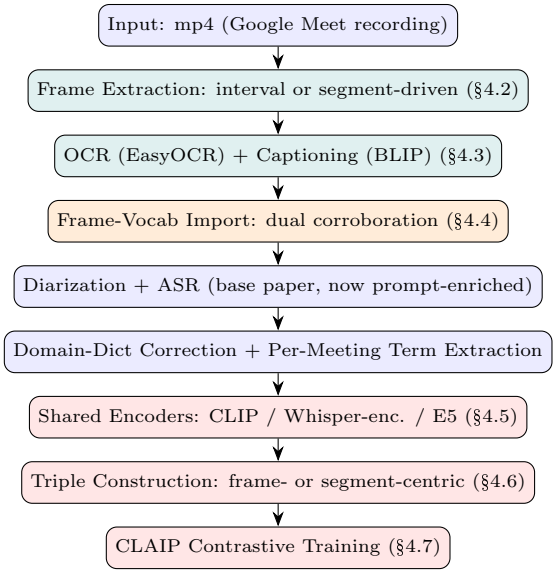


Figure 1: Extended pipeline architecture. Teal stages (image branch) run in a separate Python environment from the base paper's audio/dictionary stages (blue); orange/red stages (dual corroboration, shared embeddings, contrastive training) are this paper's central contribution. The frame-vocab import stage (§4.4) re-enriches the dictionary *before* diarization/ASR runs, closing the same within-run dictionary-expansion loop the base paper describes (Equation 2) but from the image branch.

4.2 Screen-Share Frame Extraction

We implement both instantiations of $F(a)$ (§3).

Fixed-interval sampling (F_{int}). A frame is extracted via `ffmpeg` seek-and-grab every 10 s, independent of speech activity. This is the simpler of the two designs, but a slide transition between sample points is never captured, and a long stretch of an unchanging slide is sampled repeatedly regardless of its relevance to any particular utterance.

Segment-driven sampling (F_{seg}). For every diarization segment in $D_{\text{excl}}(w)$ with duration ≥ 0.2 s (the same filter ASR applies, so sampling and transcription operate over an identical segment set), we extract

$$k_i = \min\Big(K,\ \max\big(1,\ \lceil (e_i - s_i)/\tau \rceil\big)\Big) \quad (3)$$

frames evenly spaced within $[s_i, e_i]$, where $\tau = 3.0$ s is the target sub-interval and $K = 5$ caps the per-segment sample count. A short utterance ($e_i - s_i \leq \tau$, the common case: the median segment duration in our case-study recording is 1.43 s) receives exactly one frame at its midpoint; a long monologue segment, during which the screen may have changed, receives

Algorithm 1 Segment-driven frame sampling with sequential de-duplication

Require: segments $\{(s_i, e_i)\}_{i=1}^{M}$ sorted by s_i, video a, $\tau = 3.0$, $K = 5$, hash threshold $h = 4$
Ensure: frame set with kept_p, ρ_p (represents-pointer)

cands $\leftarrow$ []; prev_hash $\leftarrow$ None; prev_id $\leftarrow$ None
for $i = 1$ to M **do**
  $k_i \leftarrow \min(K, \max(1, \lceil (e_i - s_i)/\tau \rceil))$ ▷ Eq. 3
  for $j = 1$ to k_i **do**
    $t \leftarrow s_i + (j{-}0.5)(e_i{-}s_i)/k_i$
    cands.append(grabFrame(a, t), i)
  end for
end for
for $(f, i) \in$ cands (time order) **do** ▷ sequential dedup
  $h_f \leftarrow \text{phash}(f)$
  if prev_hash = None **or** hamming$(h_f, \text{prev_hash}) \geq h$ **then**
    $\text{kept}_f \leftarrow 1$; $\rho_f \leftarrow f$; prev_hash $\leftarrow h_f$; prev_id $\leftarrow f$
  else
    $\text{kept}_f \leftarrow 0$; $\rho_f \leftarrow$ prev_id
  end if
end for

multiple. Algorithm 1 details sampling followed by global, sequential de-duplication (§3's dedup): a candidate's perceptual hash (`phash`) is compared against the *last kept* frame's hash, not merely its immediate predecessor, so a slide shown continuously across many short consecutive utterances still collapses to one kept frame rather than one per utterance.

On our 44.9-minute case-study recording, F_{seg} produces 1,231 candidate frames from 914 segments (vs. 270 for F_{int} at a 10 s interval), of which 264 survive de-duplication (vs. 92 for F_{int}) – a direct empirical consequence of sampling at the utterance's own time granularity rather than an arbitrary wall-clock interval (§6).

4.3 OCR and Image Captioning

Each kept frame is processed by two independent models. **OCR**: EasyOCR [8] (Japanese + English detection and recognition) extracts on-screen text x_p^{ocr}, retaining only detections with confidence ≥ 0.4; we selected EasyOCR over PaddleOCR specifically to avoid a second, Paddle-specific deep-learning framework dependency alongside our existing PyTorch-based stack, and over Tesseract to avoid a separately installed native binary. **Captioning**: BLIP [6] (`blip-image-captioning-base`) generates an English natural-language caption x_p^{cap}. The two signals are complementary in our setting: a document-viewer screenshot's sticky-note review comments are exact, written domain vocabulary that OCR recovers verbatim, whereas a photograph of a bridge member or a site overview is better summarized by BLIP's scene-level caption than by whatever (if any) text happens to be visible in it; the unified semantic text $x_p = x_p^{\text{ocr}} \,\|\, x_p^{\text{cap}}$ (§3) combines both for the text branch of the shared embedding space (§4.5).

4.4 Frame-Derived Vocabulary Import and Dual Corroboration

E_{frame} (§3) reuses E_{meet}'s candidate-extraction, canonicalization, and LLM-relevance-classification machinery from the base paper (Equations (canonical-meet)–(relevance) there) applied to $\{x_p^{\text{ocr}}\}$ instead of a transcript segment's text, with corroboration checked against *both* φ_{ont} and φ_{frame} (Equation 1) when later classifying audio-side candidates via E_{meet} – the result reported as src(m) in our candidate reports (§6).

Re-running this extraction on real OCR text surfaced two candidate-extraction failure modes specific to OCR, distinct from both failure modes the base paper reports for audio transcripts (naive substring collision, LLM topical-relevance misjudgment): neither arises from natural Japanese prose, since OCR text from a screenshot of stacked review comments has no word-boundary or sentence-structure cues at all.

Failure mode 1: reviewer-name/content concatenation. When several sticky notes are visually adjacent, OCR reads a reviewer's name label directly followed by their comment text with no separating character (e.g. the pattern we denote *Reviewer+genjou*, "current status," observed as a single run *Reviewer-genjou*): because the two morphemes are adjacent nouns with no intervening particle, SudachiPy's noun-run extraction (the same mechanism the base paper uses for $E_{\text{ont}}/E_{\text{meet}}$) merges them into one meaningless compound candidate. Unlike spoken audio, where a reviewer's name is almost always followed by an honorific particle that the base paper's existing honorific-suffix filter already catches, OCR text has no such separator. *Mitigation*: we reject any candidate whose first SudachiPy morpheme is tagged as a person name (part-of-speech subcategory `noun, proper-noun, person-name`), a check specific to the frame-OCR extraction path and left unapplied to E_{meet}'s audio-side candidates (whose own pilot testing, per the base paper, did not surface this pattern).

Failure mode 2: adjacent sticky-note text

duplication. The same phrase occasionally appears twice in immediately adjacent sticky notes (e.g. a damage-category label repeated across two review comments), which OCR reads as one run of exact self-repetition (e.g. *shikajuu shikajuu*, "dead load," read twice in succession). *Mitigation*: a candidate whose first half exactly equals its second half is collapsed to the half-length form before further filtering, recovering the genuine underlying term (*shikajuu*) rather than discarding or merging the meaningless doubled string.

Both mitigations were identified empirically from real candidate-extraction output (§6) and applied only to the frame-OCR extraction path, since they are specific to OCR text's lack of natural-language structure rather than a general property of Japanese noun-phrase extraction.

4.5 Shared Embedding Space Encoders

We deliberately select encoders that minimize new, additional model footprint on the shared 16 GB GPU budget (§5), reusing the project's existing model families where possible. **Image/text**: rinna's Japanese CLIP (ViT-B/16 image encoder, 12-layer Japanese text encoder) [7], chosen over an English-centric CLIP variant for direct Japanese-text compatibility with our domain vocabulary. **Audio**: rather than introduce a second, English-centric audio-pretraining model such as CLAP [5] (untested on Japanese construction-meeting speech), we reuse the encoder half of a (frozen, separately loaded) Whisper model [17] via the `transformers` library's `WhisperModel`, mean-pooling its encoder's hidden states over a 30-second window containing the target utterance – the same architecture family the base paper already uses for ASR itself (faster-whisper), minimizing new dependency and VRAM footprint. **Text**: multilingual-E5-base [9] embeds each triple's unified semantic text (OCR + caption) concatenated with its aligned audio transcript. All three encoders remain frozen throughout; only three small linear projection heads (image/audio/text $\to \mathbb{R}^{256}$) are trained (§4.7).

4.6 Triple Construction: Frame-Centric vs. Segment-Centric

Frame-centric (baseline). For each kept frame f_p, all ASR segments whose time range falls within ± 15 s of $\mathrm{ts}(f_p)$ are concatenated into one audio-text string, paired with f_p's image embedding and a Whisper-encoder embedding of the corresponding ± 15 s audio window. This yields exactly one triple per kept frame – at most 264 triples on our case-study recording – and only loosely aligns a frame with the utterances actually co-occurring with it.

Segment-centric (this paper's design). For each of the M transcribed utterance segments (914 on our case-study recording), we resolve every dense candidate frame assigned to that segment index (§4.2) to its represented, kept frame via ρ_p, de-duplicate the resulting set of represented frame identities, and emit one triple per *distinct* represented frame, pairing it with that segment's own audio embedding and transcript text (computed once per segment and cached across its triples). A segment whose shared screen never changed contributes exactly one triple; a segment during which it changed contributes one triple per distinct image shown. This guarantees $N \geq M$ (every transcribed utterance contributes at least one triple) and additionally captures genuine within-utterance visual variation that the frame-centric design cannot represent at all, since it is organized around frames rather than utterances.

4.7 CLAIP Contrastive Training

For a batch of N triples with L2-normalized projections $\{(\hat{z}_i^{\mathrm{img}}, \hat{z}_i^{\mathrm{aud}}, \hat{z}_i^{\mathrm{txt}})\}_{i=1}^{N}$, let $s_{ij}(A,B) \triangleq \hat{a}_i \cdot \hat{b}_j / \tau_{\mathrm{nce}}$ denote the scaled similarity between the i-th A-modality and j-th B-modality projection. The symmetric InfoNCE loss between two modalities A, B is

$$\mathcal{L}_{\mathrm{NCE}}(A,B) = -\frac{1}{2N}\sum_{i=1}^{N}\left[\log\frac{e^{s_{ii}}}{\sum_j e^{s_{ij}}} + \log\frac{e^{s_{ii}}}{\sum_j e^{s_{ji}}}\right], \tag{4}$$

with temperature $\tau_{\mathrm{nce}} = 0.07$. The full training objective is a weighted sum over all three modality pairs,

$$\begin{aligned}\mathcal{L} = {} & w_1\,\mathcal{L}_{\mathrm{NCE}}(\mathrm{img},\mathrm{txt}) + w_2\,\mathcal{L}_{\mathrm{NCE}}(\mathrm{aud},\mathrm{txt}) \\ & + w_3\,\mathcal{L}_{\mathrm{NCE}}(\mathrm{img},\mathrm{aud}),\end{aligned} \tag{5}$$

with per-pair weights $w_1, w_2, w_3 > 0$ left as hyperparameters.

Motivation for per-pair weighting: the standby-frame case. Triple construction (§4.6) assigns a frame to a segment purely by diarization timing, with no check on what the frame actually shows. A frame can therefore be a Google Meet *standby* tile rather than the presenter's intended shared screen: a plain black background with only the speaker's name and a small profile photograph centered on it, shown while a screen share has not yet started or has been briefly paused. For such a frame, the image embedding carries essentially no information about the segment's spoken content, so both its image-text and its image-audio similarity are, correctly, low – while the

audio-text pair, which does not depend on the image branch at all, remains comparatively strong. Left unweighted, these low-information image pairs still contribute full-strength gradient terms to $\mathcal{L}$, diluting the signal from segments whose frame genuinely does match the spoken content. Equation 5's per-pair weights give a lever to down-weight the image-paired terms (w_1, w_3) relative to the audio-text term (w_2) when such standby frames are frequent. The main results of §6 use the unweighted setting $w_1 = w_2 = w_3 = 1$ throughout; §6.5 reports a separate, seed-matched ablation testing one hand-chosen weighted setting against this unweighted baseline.

Training uses full-batch Adam (all N triples from the one recording as a single batch, since N is small enough to fit in GPU memory) with a fixed learning rate; no mini-batching or learning-rate schedule is used. Increasing N from 90 (frame-centric) to 961 (segment-centric) substantially increases the in-batch negative count (and hence task difficulty: the random-chance loss floor for each pairwise term scales with $\ln N$), and empirically induces intermittent late-training loss spikes under this fixed-learning-rate regime (§6); we mitigate this by tracking the lowest-loss epoch's parameters throughout training and restoring them at the end, rather than using whichever epoch the fixed training-length budget happens to land on.

5 Experimental Setup

5.1 Dataset

We reuse the base paper's M9 case-study recording unchanged: a 44.9-minute Google Meet video export (311 MB, h.264 video, AAC 48 kHz/stereo audio) of a bridge-repair external design review meeting, 6 detected speakers, 914 ASR-transcribed utterance segments (diarization-exclusive segments ≥ 0.2 s). We process this same recording's video stream – discarded by the base paper's audio-only preprocessing – through the image branch introduced in this paper. Reusing the same recording lets us isolate the image branch's effect without any confound from a different meeting's content or audio quality.

5.2 Frame-Extraction-Method Comparison Protocol

Because φ grows monotonically across pipeline runs (§3), naively re-running E_{frame} a second time against the *live*, already-enriched dictionary would make the second sampling strategy's candidate count look artificially smaller – many terms the first run already found would simply be excluded as "already known." To isolate the sampling *method*'s effect from this confound, we reconstruct a fixed reference dictionary as it existed before any frame-OCR merge (183 entries, $\varphi_0 \cup \varphi_{\text{ont}} \cup \varphi_{\text{meet}}$ from the base paper's own evaluation; §4.4 never rewrites φ in place, only appends, so this prefix is always recoverable from a recorded line-count snapshot) and run candidate extraction for *both* F_{int}'s and F_{seg}'s kept-frame OCR text against this same fixed reference, reporting candidate counts that are directly comparable.

Table 1: Image-branch software configuration (separate virtual environment).

Component	Version / Model
torch / torchvision	2.11.0+cu128 / 0.26.0+cu128
EasyOCR	1.7.2 (`ja+en`)
BLIP	`blip-image-captioning-base`
Japanese CLIP	`rinna/japanese-clip-vit-b-16`
Whisper (encoder only)	`openai/whisper-small`
sentence-transformers	`intfloat/multilingual-e5-base`
imagehash	4.3.2 (`phash`)

5.3 Hardware and Software Configuration

All experiments ran on the same machine as the base paper (NVIDIA GeForce RTX 4060 Ti, 16,380 MiB VRAM). The image branch runs in a separate Python virtual environment from the base paper's audio/dictionary stack (§4), summarized in Table 1; each stage's model is released from GPU memory before the next stage loads, so peak VRAM usage never requires holding more than one of these models resident at once.

5.4 Evaluation Protocol

We report, for both F_{int} and F_{seg}: candidate frames sampled, frames kept after de-duplication, total OCR character count, and candidate-term count under the fixed reference dictionary (§5.2). For dictionary expansion we report, per run: raw high-confidence merges, manual-audit outcome (genuine term vs. OCR-artifact false positive, classified by two of the authors independently reviewing every merged entry against its source frame image), and the resulting net dictionary size. For the shared embedding space we report triple count, coverage (fraction of utterance segments with ≥ 1 triple), training-loss curves, and a 7-query qualitative retrieval sanity check using the same seven high-confidence domain terms the base paper's dictionary-expansion mechanism already verified as genuine on this recording (*ateita*

Table 2: Frame-extraction density, fixed 183-term reference dictionary.

Metric	F_{int}	F_{seg}
Candidate frames sampled	270	1,231
Kept after de-duplication	92	**264**
OCR characters (total)	5,695	**13,842**
Candidate terms (fixed ref.)	55	**93**
of which new vs. other method	–	42

Table 3: Dictionary expansion via frame-derived dual corroboration.

Run	Merged	FP (audit)	Net genuine
F_{int} (v0.1.0)	29	4	25
F_{seg} (v0.2.0)	6	3	3
Total frame-derived	35	7	**28**
Audio-side E_{meet}	–	–	2
Dictionary size			183 → **213**

"steel-plate patch," *aachi-rib* "arch rib," *shita-flange* "bottom flange," *rivet*, *nakazume* "infill," *kouzou-keishiki* "structural form," *hoshuu-houhou* "repair method"): for each query, embedded via the text encoder and projection head, we report whether the nearest frame (by image projection) or audio segment (by audio projection) actually contains that term in its OCR/transcript text. We report this candidly as a qualitative, single-recording sanity check, not a statistically validated Recall@K benchmark (§7).

6 Results

6.1 Frame-Extraction Density and Candidate-Term Coverage

Table 2 and Figure 2 summarize the comparison protocol of §5.2. Segment-driven sampling (F_{seg}) nearly triples kept frames (92 to 264, +187%) and more than doubles total extracted OCR text (5,695 to 13,842 characters), improving candidate-term coverage under the identical fixed reference dictionary by 69% (55 to 93 terms); 42 of the 93 segment-driven candidates were not found by interval sampling at all. Manual inspection of these 42 segment-driven-only candidates (full list in §4.4's underlying report) confirms several genuine, previously-missed domain terms (e.g. *soujō-fushoku* "layered corrosion," *shiten-bu* "support point section," *anabiki* "bolt-hole clearance," *taiousaku* "countermeasure"), alongside a proportionally larger share of OCR noise than the coarser interval sampling produced (§7).

6.2 Dictionary Expansion via Dual Corroboration

Table 3 reports the frame-derived dictionary-expansion outcome across both sampling strategies, each applied to the *live* dictionary at the time it was run (not the fixed reference of §5.2, which exists only for the fair method comparison above). F_{int}'s run merged 29 high-confidence candidates, of which manual audit found 4 to be OCR-artifact false positives (*mama-ateita*, a sentence fragment incorporating the genuine term *ateita*; *shō-kinbou-kasho*, a truncated form of "near the bearing support, location"; *soujō-fushoku-No*, a genuine term with a photo-caption numbering suffix attached; and *kakunin-moku*, a truncated name-fragment collision), leaving 25 genuine terms merged. F_{seg}'s subsequent run against the now-further-enriched live dictionary merged 6 additional high-confidence candidates (most of the density gain's genuine new terms, Table 2, had already been captured and excluded as "known" by this point), of which 3 were OCR artifacts caught by the duplication and truncation patterns of §4.4 (*soujō-fushoku-No.*, *jı-ryō-tē-nyō*, a garbled multi-note OCR run; and *soujō-fushoku-rikkuru*, a truncated glue artifact), leaving 3 genuine terms merged (*shiten-bu*, *soujō-fushoku*, *anabiki*). Combined, dual corroboration expanded the dictionary from 183 to 213 entries (+30, of which 28 via frame OCR across both runs and 2 via the base paper's existing audio-side E_{meet} mechanism on this same recording, down from the base paper's originally reported 7 since most candidates it previously discovered post-hoc were already present in the dictionary from this paper's pre-ASR frame import, §7).

6.3 Contrastive-Triple Density and Training Stability

Table 4 and Figure 3 report the triple-construction comparison of §4.6. Segment-centric construction increases total triples more than tenfold (90 to 961) while guaranteeing full utterance coverage (914/914 segments, vs. 264 frames capping the frame-centric design's maximum possible triple count); 43 segments contributed more than one triple because the shared screen changed mid-utterance, a pattern the frame-centric design cannot represent at all. Figure 3 shows both training runs' InfoNCE loss: the frame-centric run (90 triples) converges smoothly to a near-zero loss (13.8 to 0.59 at the best checkpoint), while the segment-centric run (961 triples) converges to a substantially higher absolute loss (20.75 to 6.62 at the best checkpoint) – consistent with the larger in-batch negative count raising the random-chance floor

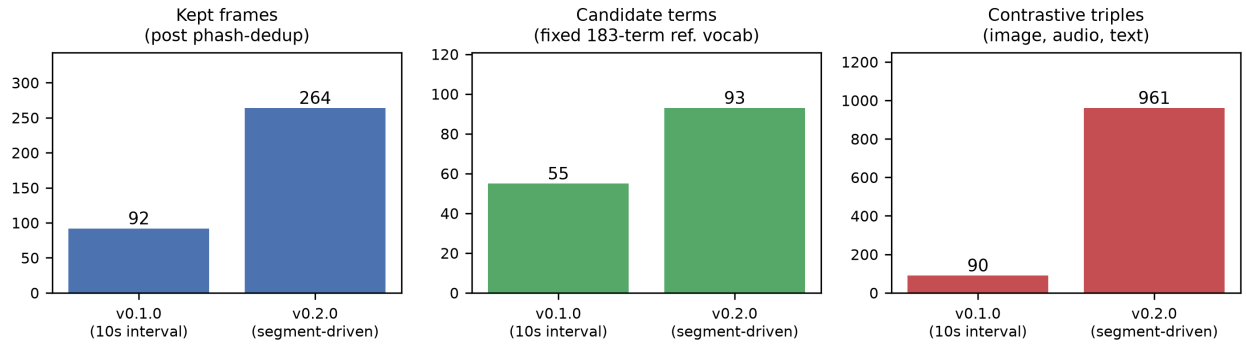


Figure 2: Frame-extraction density and contrastive-triple count, interval (F_{int}) vs. segment-driven (F_{seg}) sampling. Candidate-term counts use the fixed 183-term reference dictionary of §5.2.

Table 4: Contrastive-triple construction: frame-centric vs. segment-centric.

Metric	Frame-centric	Segment-centric
Total triples	90	**961**
Utterance coverage	–	914/914
Segments with >1 triple	–	43
Distinct images reused	–	264
Best-ckpt. InfoNCE loss	0.59	6.62

($\ln 961 \approx 6.87$ per pairwise term) – and exhibits three distinct late-training loss spikes (around epochs 140, 320, and 540 of an 800-epoch run) under our fixed learning rate, which best-checkpoint retention (§4.7) resolves without requiring a learning-rate schedule.

6.4 Cross-Modal Retrieval Sanity Check

Figure 4 reports the 7-query sanity check (§5). The frame-centric space (90 triples) correctly retrieves 2 of 7 known domain terms (*aachi-rib*, *hoshuu-houhou*); the segment-centric space (961 triples) correctly retrieves 4 of 7, adding *ateita* and *shita-flange* while still missing *rivet*, *nakazume*, and *kouzou-keishiki*. We report this as a qualitative, single-recording proof of concept: with only 7 queries and one video's embedding space, these counts should not be read as a precisely estimated retrieval accuracy (§7).

6.5 Loss Weighting for the Standby-Frame Hypothesis

To test whether the per-pair loss weighting motivated in §4.7 is actually beneficial, we trained two projection heads on the same 961 segment-centric triples

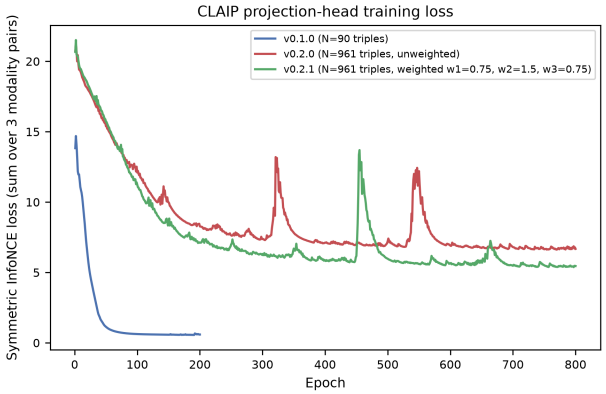


Figure 3: CLAIP projection-head InfoNCE training loss. The segment-centric run's larger in-batch negative count (961 vs. 90 triples) raises both the achievable loss floor and the visible late-training instability under fixed-learning-rate Adam (§7.3). The third curve (v0.2.1, weighted) is the standby-frame-motivated weighted objective of §6.5 ($w_1 = 0.75, w_2 = 1.50, w_3 = 0.75$), trained on the same 961 triples with the same random seed as a controlled unweighted rerun (not separately plotted here; see Table 5).

with a fixed random seed (so the two runs differ only in w_1, w_2, w_3, not in initialization): an unweighted control ($w_1 = w_2 = w_3 = 1$) and a hand-chosen weighted setting ($w_1 = 0.75$, $w_2 = 1.50$, $w_3 = 0.75$, summing to the same total weight as the unweighted case). Table 5 reports the outcome. The weighted run reaches a lower best-checkpoint loss (5.38 vs. 6.69) and shows one late-training loss-spike event instead of two over the same 800 epochs (Figure 3). Because $w_1, w_3 < 1 < w_2$ changes the objective's

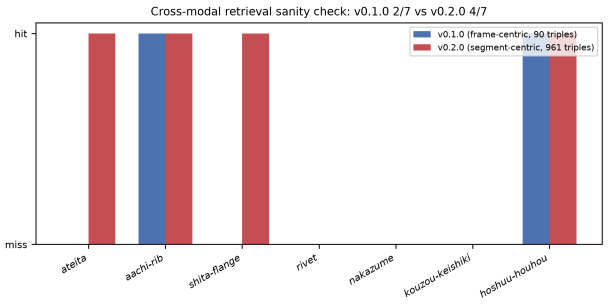


Figure 4: Cross-modal retrieval sanity check, 7 known high-confidence domain terms (romanized). Segment-centric training (961 triples) nearly doubles the hit rate observed with frame-centric training (90 triples).

nominal scale, we additionally re-scored both runs' best checkpoints under the common, *unweighted* sum $\mathcal{L}_{\text{NCE}}(\text{img}, \text{txt}) + \mathcal{L}_{\text{NCE}}(\text{aud}, \text{txt}) + \mathcal{L}_{\text{NCE}}(\text{img}, \text{aud})$ to separate a genuine per-pair improvement from this scale effect: on this common footing, the audio-text pair improves by 19.8% (0.803 to 0.644), while the image-text and image-audio pairs are essentially unchanged (within 1.5%, i.e. not measurably degraded despite their lower weight). The 7-query retrieval hit rate (§5) is unchanged at 4/7 for both runs.

7 Discussion

7.1 Why Utterance-Aligned Sampling Density Matters

The base paper's processing-time analysis found that per-segment invocation overhead means high-turn-density meetings cost disproportionately more to process; this paper's frame-density results generalize the same underlying principle from *cost* to *coverage*: sampling at a granularity tied to the unit of downstream analysis (the diarized utterance) rather than an arbitrary wall-clock interval recovers more of the content that matters for that analysis, whether the cost being optimized is processing time (base paper) or candidate-term/training-data coverage (this paper). A fixed interval is both too coarse (slide transitions between sample points are simply missed, §4.2) and too indiscriminate (many redundant samples of an unchanging slide, later discarded by de-duplication) relative to sampling that follows the audio's own segmentation.

7.2 A Third Candidate-Extraction Failure-Mode Family

The base paper reports two distinct audio-side candidate-extraction failure-mode families: naive substring collision at the morphological level, and LLM topical-relevance misjudgment at the semantic level. This paper's two OCR-specific failure modes (§4.4) constitute a third family, arising neither from Japanese morphology nor from LLM judgment, but from OCR text's complete lack of natural-language structure (no word boundaries where there would be whitespace in a Western script, no particles, no sentence punctuation separating visually adjacent but semantically unrelated screen content). The generalizable lesson across all three families is the same as the base paper's for morphology specifically: *any* automated vocabulary-candidate-extraction mechanism must be audited against the specific structural properties of its actual input text, not assumed safe merely because a similar mechanism worked safely on a different text source (audio transcript vs. external ontology vs., now, OCR). Notably, the proportion of false positives among frame-derived merges (7 of 35, 20%) is substantially lower than the base paper's reported audio-side per-meeting false-positive rate before mitigation (11 of 14, 79%, on an earlier meeting, improving to 0 of 7 only after four dedicated mitigations) – consistent with frame OCR's candidates being exact, verbatim written text rather than a speech-recognition hypothesis layered on top of an acoustic model's own uncertainty, even though OCR introduces its own distinct noise sources.

7.3 Training Stability at Increased Triple Count

The qualitative difference between the two training curves in Figure 3 – monotonic near-convergence at 90 triples versus a higher loss floor with intermittent spikes at 961 – is, to our knowledge, not previously reported for this small-scale, single-recording contrastive training regime, where triple count is set entirely by how one real meeting's content happens to be structured rather than by a deliberately chosen dataset size. We attribute the increased instability to the larger in-batch negative count interacting with a fixed learning rate under full-batch Adam (no mini-batching, since all triples fit in GPU memory): as the embedding space sharpens, a given gradient step can more easily overshoot a now-narrower optimum, producing the isolated spikes visible in Figure 3 around epochs 140, 320, and 540. Best-checkpoint retention is a minimal, zero-additional-hyperparameter

Table 5: Standby-frame-motivated loss weighting: seed-matched unweighted vs. weighted ($w_1 = 0.75, w_2 = 1.50, w_3 = 0.75$), same 961 triples.

Metric	Unweighted	Weighted	Δ
Best-ckpt. loss (as trained)	6.69	**5.38**	-19.6%
Final-epoch loss	7.08	**5.46**	-22.9%
Late-training spike events	2	**1**	–
Per-pair loss, re-scored unweighted (common basis):			
image-text	2.936	2.979	+1.5%
audio-text	0.803	**0.644**	**-19.8%**
image-audio	2.996	3.020	+0.8%
Retrieval hit rate (7 queries)	4/7	4/7	–

mitigation; we did not evaluate whether a learning-rate schedule would avoid the spikes altogether, or whether they recur at a different triple count, and flag both as natural next steps once multi-recording training (§8) makes mini-batching necessary regardless.

7.4 Loss Weighting Mitigates Training Instability, but Mostly via Audio-Text

The weighted run's lower raw loss and halved spike count (§6.5) might at first look like a general sharpening of the shared embedding space, but the per-pair re-scoring in Table 5 shows this is not quite the right reading: nearly all of the measured improvement is concentrated in the audio-text pair, the one pair we up-weighted ($w_2 = 1.50$) and whose hypothesis-driven motivation (§4.7) was precisely that it should remain reliable even when a frame is uninformative. The two down-weighted, image-involving pairs ($w_1 = w_3 = 0.75$) are not degraded, but neither do they improve – they are, as expected, simply getting slightly less gradient signal per step. We read this as mild, directionally consistent support for the standby-frame hypothesis (up-weighting the modality pair least affected by an uninformative image does help that pair, without visibly harming the others) rather than evidence that this specific weight choice meaningfully improves the trained space's overall cross-modal alignment: the unchanged 4/7 retrieval hit rate (Table 5) is consistent with this more modest interpretation. A sharper test of the hypothesis would require first measuring how many triples in a recording actually contain a standby frame (not done here, §7.2) and comparing weighted training with and without those specific triples present, rather than applying a single hand-chosen weight triple uniformly to all 961 triples regardless of whether each one's image is a standby tile.

7.5 Limitations

We report the following limitations candidly, extending the base paper's own Limitations subsection. First, both the frame-extraction-method comparison and the CLAIP training/retrieval results are a *single-recording* case study; the dictionary-expansion false-positive rates and training-stability observations of §7.2–§7.3 are informative single-instance evidence, not statistically estimated rates, matching the base paper's own small-sample caveat for its audio-side dictionary-expansion evaluation. Second, the 7-query retrieval sanity check (§6) uses a closed, 264-to-961-row candidate set drawn from the same single recording it was trained on; it demonstrates that the mechanism functions as designed, not a generalizable retrieval accuracy, and future work should evaluate on a held-out recording once multiple recordings are available. Third, the sub-sampling interval ($\tau = 3.0\,\mathrm{s}$) and per-segment cap ($K = 5$) in Equation 3 are empirical choices tuned to this recording's statistics (median segment duration 1.43 s) and untested on meetings with substantially different speech-turn or screen-share-change cadence. Fourth, the external `rinnakk/japanese-clip` repository referenced by the model's own documentation was unavailable (HTTP 404) at the time of this work, requiring us to install an alternative community fork and apply two compatibility patches for API drift against current `transformers`/ `huggingface_hub` releases; this is a reproducibility risk inherent to depending on an actively-evolving third-party ecosystem rather than a methodological limitation of the approach itself, but future replications should expect to re-verify this dependency chain. Fifth, the InfoNCE temperature ($\tau_{\mathrm{nce}} = 0.07$), projection dimensionality (256), and training-epoch budget (800 for the segment-centric run) were fixed by convention rather than tuned via a hyperparameter sweep. Sixth, the loss-weighting ablation (§6.5) tests a single hand-chosen weight triple on a single recording with a single random seed; we did not sweep w_1, w_2, w_3 systematically, nor did we

measure how many of the 961 triples' frames are actually standby tiles (§7.4), so we cannot attribute the observed audio-text improvement specifically to standby frames being down-weighted versus a more general effect of shifting relative pair weights; detecting standby frames automatically (e.g., from their near-uniform black background) and comparing training with and without them present is a prerequisite for a more direct test of the hypothesis.

7.6 Practical Implications

For practitioners, the central practical finding is that *how* a fixed amount of screen-share video is sampled matters as much as *whether* it is used at all: aligning frame sampling to the same utterance segmentation already computed for diarization, rather than an independent wall-clock interval, is a low-cost change (no additional model, only a different set of `ffmpeg` seek timestamps) that measurably improves both dictionary-expansion coverage and contrastive-training data density. The dual-corroboration mechanism (§4.4) further lets an organization's own in-meeting screen content – design drawings, inspection photographs, sticky-note review annotations – directly improve same-run ASR vocabulary bias, requiring no external knowledge source beyond the recording itself, complementary to the base paper's ontology-corroborated bulk import.

8 Conclusion

8.1 Concluding Remarks

We extended an on-premise construction-meeting ASR pipeline's dictionary-correction mechanism with a second, independent corroboration source – screen-share frame OCR, available before speech recognition runs – and built a three-modality shared embedding space via CLAIP-style contrastive learning over the same recording's own utterance-aligned data. Our central empirical finding is that aligning frame sampling to the diarization segmentation already computed for ASR, rather than sampling at a fixed wall-clock interval, substantially improves both candidate-term coverage (+69% under a controlled, fixed-reference-vocabulary comparison) and contrastive-training data density (90 to 961 triples, full utterance coverage), which together nearly double a qualitative cross-modal retrieval sanity check (2/7 to 4/7). We additionally reported a third family of candidate-extraction failure modes specific to OCR text, and an empirical contrastive-training-stability finding as triple count scales by an order of magnitude on a single recording. Motivated by the standby-frame case – a Google Meet participant tile entering a triple in place of genuine shared-screen content – we introduced a per-modality-pair loss weighting and, in a seed-matched ablation, found that one hand-chosen weighted setting sharpens the audio-text pair substantially (-19.8%) without degrading the image-paired terms and roughly halves the observed late-training spike count, though it leaves the qualitative retrieval hit rate unchanged – a safe but modest refinement on this evidence, not a demonstrated fix. As with the base paper, we report all results candidly, including a single-recording case study's inherent limitations (§7).

8.2 Future Work

In the near term, we plan to (i) extend this evaluation to multiple recordings, both to obtain statistically meaningful dictionary-expansion false-positive rates and retrieval accuracy (§7) and to make mini-batched, learning-rate-scheduled contrastive training necessary and evaluable (§7.3); (ii) crop the shared-screen region of a frame before computing its perceptual hash, rather than hashing the full frame including the Google Meet participant-gallery tile, to test whether de-duplication becomes more precise; (iii) evaluate a learning-rate schedule (warmup/decay) as an alternative or complement to best-checkpoint retention; and (iv) detect standby frames automatically (e.g., a near-uniform black background with a single centered name/photo tile), measure their actual frequency among the 961 triples, and use that detector to compare training with and without the affected triples present, or to sweep Equation 5's per-pair weights w_1, w_2, w_3 across multiple seeds and settings rather than the single hand-chosen triple evaluated in §6.5, so that any further gain can be attributed specifically to standby frames rather than to a general shift in relative pair weights. Beyond these near-term items, this extension is a direct step toward the project's own Concept Note vision of CLAIP-based cross-modal meeting search: with a trained shared embedding space in place, a natural next step is a query interface over the corroborated vector store – e.g., FAISS-indexed nearest-neighbor search over $\pi_{\text{img}}, \pi_{\text{aud}}, \pi_{\text{txt}}$ projections – letting an engineer search a meeting archive by natural-language query across audio, screen content, and text simultaneously. More broadly, the dual-corroboration principle demonstrated here – that an organization's own in-meeting artifacts (here, screen-share content) can directly verify and expand a domain dictionary without depending solely on an external knowledge source –

suggests further independent evidence sources worth corroborating against in future work, such as shared design documents or chat logs exchanged during the same meeting, extending the same dual-corroboration mechanism to additional in-meeting modalities beyond audio and screen-share video.

Looking further ahead, this bridge-repair design-review case study is best read as a first domain instance of the broader Meeting Intelligence goal introduced in §1: using an organization's own meeting video, rather than an external corpus, as the raw material for continuously corroborated, specialized-domain ontology construction. Having validated the dual-corroboration and segment-aligned contrastive-triple mechanisms within one construction sub-discipline (bridge structural repair), a natural cross-sectional extension is to the many other specialized engineering departments an infrastructure-services organization such as ours maintains across the broader construction field – roads and railways, rivers/dams and erosion control (*sabo*), architecture and parks, and environment/energy – each with its own damage-cause vocabulary, review-meeting conventions, and screen-share artifact types, but sharing the same underlying pipeline. We envision this domain adaptation as populating an agentic mesh [18] of department-specific ontology-corroboration and retrieval agents, each independently verified against its own domain evidence yet interoperable through a shared CLAIP-style embedding space and cross-domain workflow-composition mechanisms [19], so that a query spanning multiple specialized departments (e.g., a river-crossing bridge repair touching both structural and hydraulic domains) can be routed to, and corroborated by, the relevant department-specific agents without re-deriving a shared ontology from scratch.

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